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Gear Contact Model and Loaded Tooth Contact Analysis of a Three-Dimensional, Thin-Rimmed Gear

This paper performs loaded tooth contact analysis of a three-dimensional, thin-rimmed gear (3DTRG) by presenting a method that combines the mathematical programming method with the three-dimensional, finite element method (3DFEM). Also, a face-contact and whole gear deformation model is used for the 3DTRG. 3DFEM programs for the contact analysis and strength calculation of the 3DTRG are developed successfully in a personal computer. By using this program, 3D tooth load distributions, tooth root strains and the tooth contact pattern of the 3DTRG are obtained. Calculation results are proved to be correct by experiments. [DOI: 10.1115/1.1485290]

1 Introduction

Gears used in aircraft and spaceships must have very thin-walled structures to reduce flying weight. This kind of gear is usually called the thin-rimmed gear (TRG) by gear researchers. In recent years, this kind of gear has been finding wider applications in cars and other common machines to meet lightweight need. But strength design problems, especially stress calculation problems for this kind of gear have not been solved yet. This paper presents a method and program for the gear designers to calculate stresses of this kind of gear. As the first step of strength study of the 3DTRG, tooth loads are investigated.

A good deal of research [1–4] on the strength design problem of a two-dimensional, thin-rimmed gear can be found, but very little documentation is available for the 3DTRG. Abiy and Graziano [5,6] studied 3DTRG stress calculations. The partial tooth deformation models that were used for a solid gear stress analysis were also used for the 3DTRG. The difference between the partial tooth deformation model and the whole gear deformation model shall be discussed in this paper.

It is often assumed that tooth loads are distributed only on the geometric contact line when they are calculated for the solid gears. This is called line-contact model in this paper. The reality is that gear contact is on a face around the geometric contact line because of tooth surface deformation. This paper suggests a face-contact model for contact analysis of the 3DTRG. The difference between the line-contact and the face-contact is also discussed.

Conry and Serireg [7–8] presented a mathematical programming method and used it to calculate the tooth loads of the solid gears successfully. Ji [9] combined this method with the two-dimensional, finite element method. The author has developed Conry and Ji's method into a three-dimensional finite element method through many years of research. The method calculates the tooth load distribution of 3DTRG. Similar research was conducted by Prabhu and Houser [10], but the gear model and the treatment of contact points are different.

2 Face-Contact Model of Gears

Engagement of a pair of spur gears ①, ② on the geometric contact line is shown in Fig. 1(*b*). Figure 1(*a*) is a 3D view of a tooth surface of the gear ①. In Fig. 1(*a*), loaded tooth contact is assumed to be on a reference face with a width that is a part of the tooth surface. The geometrical contact line of the gear is located on the center of the reference face. Many points on the reference face of gear ① are made artificially. They are called reference points as shown in Fig. 1(*a*). Tooth contact on the reference face of gear ① is replaced by the contact on the reference points of gear ①. Contact reference points on tooth surface of gear ② are looked for by a geometric method according to the positions of the reference points on gear ①.

3 Principles of Contact Analysis

3.1 Deformation Compatibility Relationship. In Fig. 2, ① and ② are one pair of elastic bodies which will contact each other under the action of an external force **P**. The contact of this pair of elastic bodies is handled as the contact of many point pairs on both supposed contact surfaces of ① and ② like the gear's contact as shown in Fig. 1. These contact point pairs are expressed as (1-1′), (a-a′), (k-k′), (j-j′), (b-b′), ⋯, and (n-n′). $\boldsymbol{\varepsilon}_\mathbf{k}$ is the clearance between a optional contact point pair (k-k′) on the two contact surfaces before contact. $\mathbf{F}_\mathbf{k}$ is the contact force between **k** and **k′** in the direction of the common normal line of the point pair (*k-k′*) when **k** contacts with **k′** under the load **P** (It is assumed that all the common normal lines of the contact point pairs are approximately along the same direction of the load **P** in this paper because a contact width is usually very narrow.) $\boldsymbol{\omega}_\mathbf{k}$, $\boldsymbol{\omega}_{\mathbf{k}'}$ are deformations of the points **k** and **k′** in the direction of the force $\mathbf{F}_\mathbf{k}$ after contact. $\boldsymbol{\delta}_\mathbf{0}$ is initial minimum clearance between ① and ② and $\boldsymbol{\delta}$ is the relative displacement of the points $\mathbf{O}_1$ and $\mathbf{O}_2$ (the loading points in Fig. 2(*b*)).

For the optional contact point pair (k-k′), if (k-k′) contacts, $(\boldsymbol{\omega}_\mathbf{k}+\boldsymbol{\omega}_{\mathbf{k}'}+\boldsymbol{\varepsilon}_\mathbf{k})$ the amount of the deformations and clearance on the point pair (k-k′), shall be equal to the relative displacement quantity $\boldsymbol{\delta}$, and if (k-k′) doesn't contact, $(\boldsymbol{\omega}_\mathbf{k}+\boldsymbol{\omega}_{\mathbf{k}'}+\boldsymbol{\varepsilon}_\mathbf{k})$ shall be greater than $\boldsymbol{\delta}$. Equations (1) and (2) can express these relationships. If the relationships are expressed with one equation, Eq. (3) is obtained.

$$\boldsymbol{\omega}_\mathbf{k}+\boldsymbol{\omega}_{\mathbf{k}'}+\boldsymbol{\varepsilon}_\mathbf{k}-\boldsymbol{\delta}>0 \quad \text{(Not contact)} \tag{1}$$

$$\boldsymbol{\omega}_\mathbf{k}+\boldsymbol{\omega}_{\mathbf{k}'}+\boldsymbol{\varepsilon}_\mathbf{k}-\boldsymbol{\delta}=0 \quad \text{(Contact)} \tag{2}$$

$$\boldsymbol{\omega}_\mathbf{k}+\boldsymbol{\omega}_{\mathbf{k}'}+\boldsymbol{\varepsilon}_\mathbf{k}-\boldsymbol{\delta}\geqq 0 \quad (k=1,2,\ldots,n) \tag{3}$$

Because only elastic deformations are considered in the contact analysis, $\boldsymbol{\omega}_\mathbf{k}$ and $\boldsymbol{\omega}_{\mathbf{k}'}$ can be expressed with Eq. (4) by using deformation influence coefficients $\mathbf{a}_\mathbf{kj}$ and $\mathbf{a}_{\mathbf{k}'\mathbf{j}'}$.

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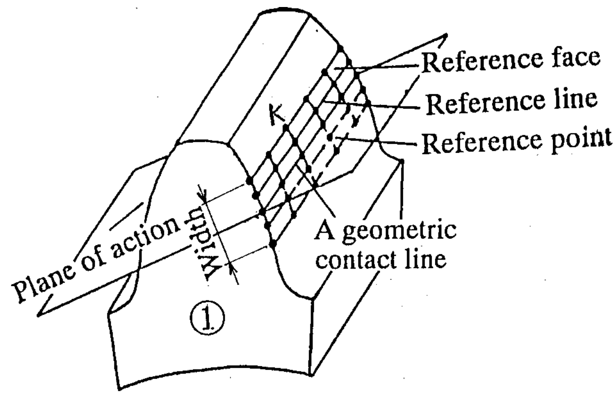
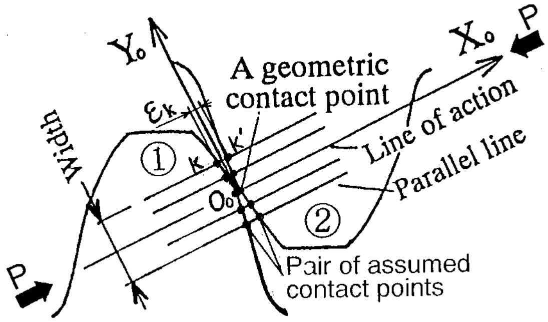
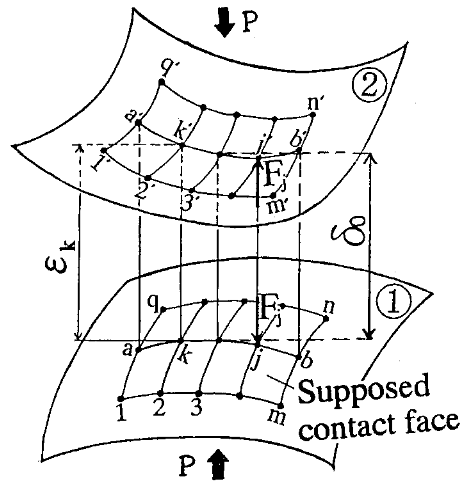


(a)

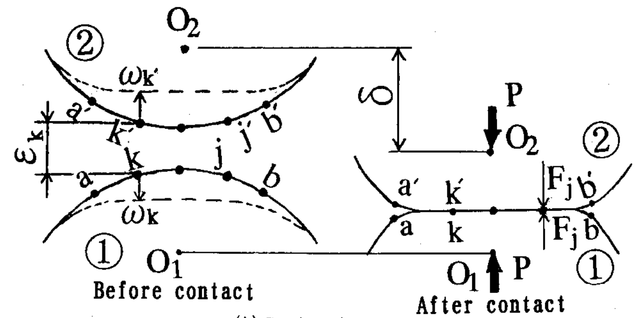


(b)

Fig. 1 Face-contact model of gears



(a) Three-dimensional view



(b) Section view

Fig. 2 Contact model of elastic bodies

$$\boldsymbol{\omega}_{\mathbf{k}}=\sum_{j=1}^{n} a_{kj}\mathbf{F_j}, \quad \boldsymbol{\omega}_{\mathbf{k'}}=\sum_{j=1}^{n} a_{k'j'}\mathbf{F_j} \tag{4}$$

Where $\mathbf{F_j}$ is the contact force between the point pair (j-j′). When Eq. (4) is substituted into Eq. (3), Eq. (5) can be obtained. Equation (6) is the matrix expression of Eq. (5).

$$\sum_{i=1}^{n} (a_{kj}+a_{k'j'})\mathbf{F_j}+\varepsilon_k-\delta \geqq 0 \tag{5}$$

$$[S]\{\mathbf{F}\}+\{\boldsymbol{\varepsilon}\}-\delta\{\mathbf{e}\}\geqslant\{0\} \tag{6}$$

Where

$[S] \;=\; [S_{kj}]=[a_{kj}+a_{k'j'}]$
$\{\mathbf{F}\} \;=\; \{F_1,\, F_2\cdots F_k\cdots\, F_n\}^T$
$\{\boldsymbol{\varepsilon}\} \;=\; \{\varepsilon_1\ \varepsilon_2\ \cdots\varepsilon_k\cdots\ \varepsilon_n\}^T$
$\{\mathbf{e}\} \;=\; \{1\ 1\ \cdots 1\cdots\ 1\}^T$
$\{0\} \;=\; \{0\ 0\ \cdots 0\cdots\ 0\}^T$
$(k,j \;=\; 1,2,\cdots n;\ k',j'=1',2',\cdots,n')$

3.2 Force Equilibrium Relationship. Also, it is because the external force **P** is equal to the sum of all the contact force $\mathbf{F_j}$ ($j=1$ to n), Eq. (7) is obtained. Equation (8) is the matrix form of Eq. (7).

$$\overset{n}{\underset{j=1}{\mathbf{F_j}}}=P \tag{7}$$

$$\{\mathbf{e}\}^T\{\mathbf{F}\}=P \tag{8}$$

Equations (6) and (8) are the constrain conditions to identify which points pair contacts or not. Contact problem of the pair of elastic bodies ① and ② can be stated as how to look for the contact load $\mathbf{F_j}$ ($j=1,2,3,\cdots,n$) that meets Eqs. (6) and (8) under the condition of knowing the deformation influence coefficients $\mathbf{a}_{kj}$, $\mathbf{a}_{k'j'}$, inertial clearance $\boldsymbol{\varepsilon}_k$ and the external force **P** in advance.

3.3 Modeling of the Contact Problem with the Mathematical Programming Method. The mathematical programming method is used to solve the contact problem by building a mathematical programming model of Modified Simplex Method [11–15].

Since Eq. (6) is an inequality constraint equation that may be strictly positive or identically zero, it should be transformed into an equality constraint equation by introducing a so-called slack variable (a positive variable) according to the principle of the mathematical programming [14]. Then Eqs. (9) and (10) can be obtained.

$$[S]\{\mathbf{F}\}+\{\boldsymbol{\varepsilon}\}-\delta\{\mathbf{e}\}-[I]\{\mathbf{Y}\}=\{0\} \tag{9}$$

or

$$-[S]\{\mathbf{F}\}+\delta\{\mathbf{e}\}+[I]\{\mathbf{Y}\}=\{\varepsilon\} \tag{10}$$

Where

$\{\mathbf{Y}\} \;=\; \{Y_1Y_2\cdots Y_k\cdots Y_n\}^T$ (slack variables)
$[\mathbf{I}] \;=\;$ unit matrix of $n\times n$

Equations (8) and (10) only are two equality constraint equations. An objective function is needed when to perform mathematical programming. According to the principle of Modified Simplex Method [14], the objective function can be given by in-

Table 1 The Simplex tableau

F_1	F_2	…	F_{n-1}	F_n	δ	Y_1	Y_2	…	Y_{n-1}	Y_n	Z_1	Z_2	…	Z_{n-1}	Z_n	Z_{n+1}	
$-S_{11}$	$-S_{12}$	…	$-S_{1n-1}$	$-S_{1n}$	+1	1					1						ε_1
$-S_{21}$	$-S_{22}$	…	$-S_{2n-1}$	$-S_{2n}$	+1		1					1					ε_2
…	…	…	…	…	…			…					…				
…	…	…	…	…	…				…					…			
$-S_{n1}$	$-S_{n2}$	…	$-S_{nn-1}$	$-S_{nn}$	+1					1					1		ε_n
1	1		1	1												1	P

troducing some positive variables X_{n+1}, $X_{n+2}, \ldots X_{n+n}$, X_{n+n+1} (usually called artificial variables) to every constraint equation. Then the mathematical programming model of the contact problem can be made as follows.
Objective function:

$$Z = X_{n+1} + X_{n+2} + \cdots + X_{n+n} + X_{n+n+1} \tag{11}$$

Constraint conditions:

$$-[S]\{\mathbf{F}\} + \delta\{\mathbf{e}\} + [I]\{\mathbf{Y}\} + [I]\{\mathbf{Z}'\} = \{\boldsymbol{\varepsilon}\} \tag{12}$$

$$\{\mathbf{e}\}^T\{\mathbf{F}\} + X_{n+n+1} = P \tag{13}$$

Where

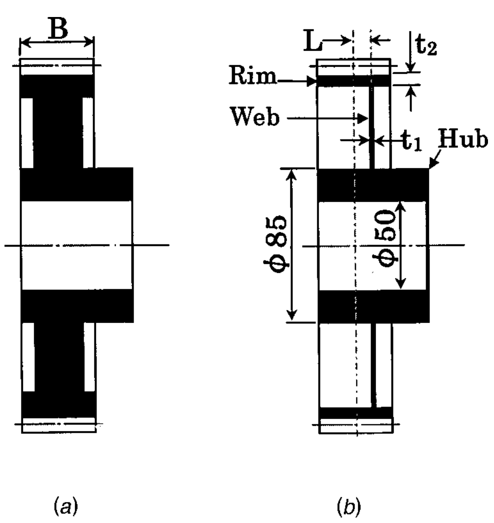
$$\begin{aligned}\{\mathbf{Z}'\} &= \{X_{n+1}, X_{n+2}, \cdots, X_{n+n}\}^T\\ \{\mathbf{F}\} &= \{F, F_2 \cdots F_k \cdots F_n\}^T\\ \{\mathbf{Y}\} &= \{Y_1, Y_2 \cdots Y_k \cdots Y_n\}^T\\ \{\boldsymbol{\varepsilon}\} &= \{\varepsilon_1\ \varepsilon_2 \cdots \varepsilon_k \cdots \varepsilon_n\}^T\end{aligned}$$

$F_k \geqq 0$, $Y_k \geqq 0$, $\varepsilon_k \geqq 0$, $\delta \geqq 0$
$X_{n+m} \geqq 0$, $m = 1,2,\cdots,n+1$

The contact force $\mathbf{F_j}$ and the relative displacement quantity $\boldsymbol{\delta}$ can be calculated by minimizing the objective function $\mathbf{Z}$ under the constraint conditions of Eqs. (12) and (13). This procedure is performed by the Modified Simplex Method [14–15] and using the simplex tableau constructed in the way as shown in Table 1.

4 Loaded Tooth Contact Analysis

4.1 Gear Structures. Gears used as the research objects are shown in Fig. 3. In Fig. 3, (*a*) is a thin-rimmed spur gear, (*b*) is a solid spur gear. Gear parameters are given in Table 2 and structure dimensions are shown in Table 3. In order to minimize effects of machining errors and assembly misalignments, the gears are ground and the gearbox is made carefully. Also, friction of the contact tooth surfaces is ignored in the contact analysis because gear transmission has a very high efficiency (about 95%～98%).



(*a*) (*b*)

Fig. 3 Structure of a pair of spur gears

4.2 Finite Element Model. Super-parametric hexahedron solid element, which has 8 nodes on the corner and 3 nodes inside the element, is used. The 3 nodes inside the element improved the calculating accuracy of a usual 8-node element (which only has nodes on the corner). It was found by many calculations of the author that this element is very suitable for contact analysis. This is because even if the element becomes very long and narrow, it still has a good calculating accuracy and stable solutions. Shape functions are shown in the following. The detailed information can be found in references [16–18].

$$\begin{cases} N_1(\xi,\eta,\zeta) = (1-\xi)(1-\eta)(1+\zeta)/8\\ N_2(\xi,\eta,\zeta) = (1+\xi)(1-\eta)(1+\zeta)/8\\ N_3(\xi,\eta,\zeta) = (1+\xi)(1+\eta)(1+\zeta)/8\\ N_4(\xi,\eta,\zeta) = (1-\xi)(1+\eta)(1+\zeta)/8\\ N_5(\xi,\eta,\zeta) = (1-\xi)(1-\eta)(1-\zeta)/8\\ N_6(\xi,\eta,\zeta) = (1+\xi)(1-\eta)(1-\zeta)/8\\ N_7(\xi,\eta,\zeta) = (1+\xi)(1+\eta)(1-\zeta)/8\\ N_8(\xi,\eta,\zeta) = (1-\xi)(1+\eta)(1-\zeta)/8\\ N_9(\xi,\eta,\zeta) = 1-\xi^2\\ N_{10}(\xi,\eta,\zeta) = 1-\eta^2\\ N_{11}(\xi,\eta,\zeta) = 1-\zeta^2 \end{cases} \tag{14}$$

Where $N_1 \sim N_8$ is the shape function of the nodes at the corner of the element, $N_9 \sim N_{11}$ is the shape function of the nodes inside the element.

The finite element model and mesh dividing of the gears is shown in Fig. 4. A four-tooth FEM model as shown in the upper part of Fig. 4(*a*) is used for the solid gear. A whole gear FEM model as shown in the lower part of Fig. 4(*a*) is used for the 3DTRG.

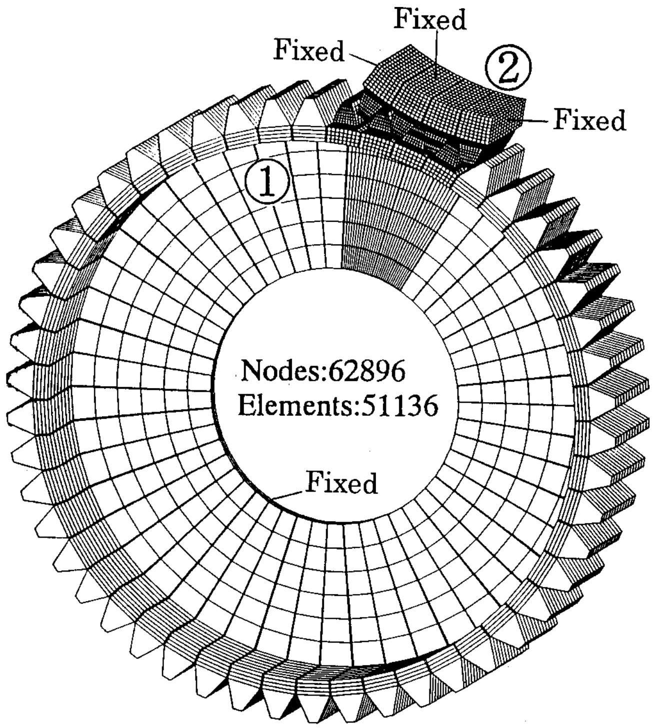
4.3 Positions of Contact Point Pairs. In Fig. 1(*a*), an optional contact reference point **k** on the reference face of gear① is made first, then a section which passes through the point **k** and is parallel to the end face of gear ①, ② is made. This section is

Table 2 Structure of a pair of spur gears

Standard involute spur gear		
Number of teeth	Z1=Z2	50
Module	m	4
Pressure angle	α_0	20°
Contact ratio		1.75
Addendum modification Coefficient		0.0
Gear accuracy		JIS 1st
Material		SCM415

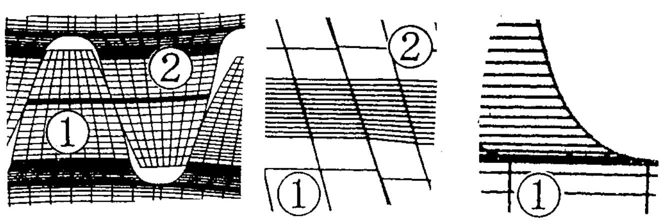
Table 3 t_1, t_2, L and B (mm)

Gear	t_1	t_2	L	B
A	26	12	0	40
B	4	5	11	40



(a) 3D view of gear contact model

(a)

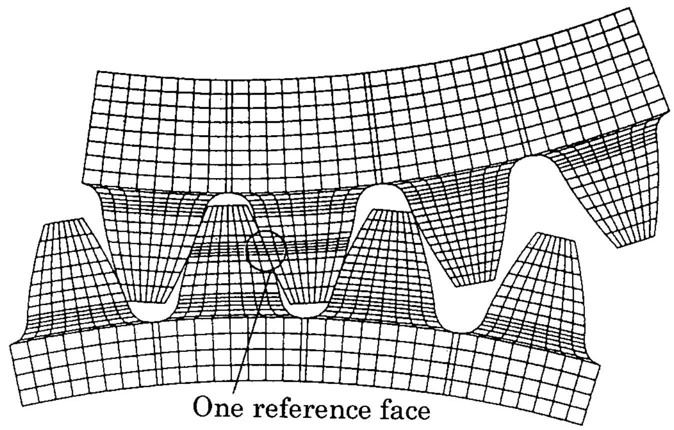


(b) Contact teeth (c) Contact area (d) Tooth root

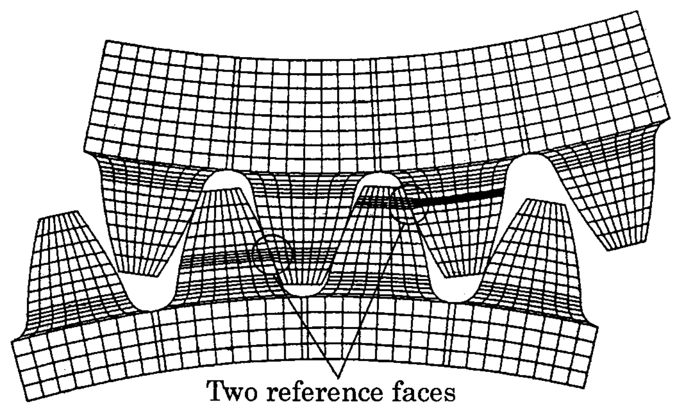
Fig. 4 FEM contact model

shown in Fig. 1(*b*). In Fig. 1(*b*), the tooth profile of gear①, ②, the line of action and the geometrical contact point $\mathbf{O_0}$ are shown at the same time. The line of action is the common normal line of the contact point $\mathbf{O_0}$. A line that passes through the point **k** and is parallel to the line of action is made. This parallel line is regarded as the common normal line of point **k** approximately. This approximation is acceptable in engineering because the contact width of one pair of elastic bodies is usually very narrow. The parallel line crosses the tooth profile of gear ② at point $\mathbf{k}'$ as shown in Fig. 1(*b*). Then point $\mathbf{k}'$ is assumed to be the contact pair point of the point **k**. In this way, all the reference points on the reference face of gear ① can find their responsive contact pair points on the tooth surface of gear② geometrically. Clearance $\boldsymbol{\varepsilon}_{\mathbf{k}}$ is obtained by calculating the distance between the point **k** and $\mathbf{k}'$.

The contact width (**Width** in Fig. 1) should be wider than the real contact width of the 3DTRG in order to get a real contact load distribution. But too big of a **Width** shall increase FEM meshes and calculating time. So a suitable **Width** should be given before the calculation. But it is difficult thing to estimate the **Width**. This paper suggests an approximate value obtained through many calculations as follows.



(a) One tooth engagement



(b) Two teeth engagements

Fig. 5 Tooth contact model and FEM meshes

$$\mathbf{Width} = (1.85 \sim 2)A \tag{15}$$

Where, A is the contact width of a pair of solid gears with the same gear parameters as 3DTRG. A is calculated by Hertzian's formula.

4.4 Deformation Influence Coefficients. Deformation influence coefficient matrix $[a_{kj}]$ of the 3DTRG (gear①) is calculated by FEM and using the whole gear FEM model shown in the lower part of Fig. 4(*a*). A unit force is exerted on the point **k** (see Fig. 1(*a*)) along the line of action, then deformations of all the reference points on tooth contact reference face of gear① in the

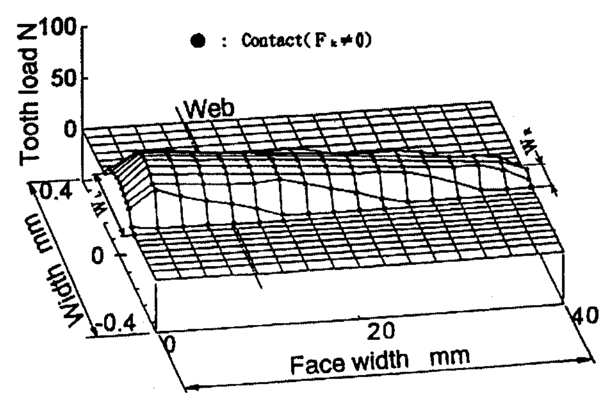


Fig. 6 Tooth load of one tooth engagement

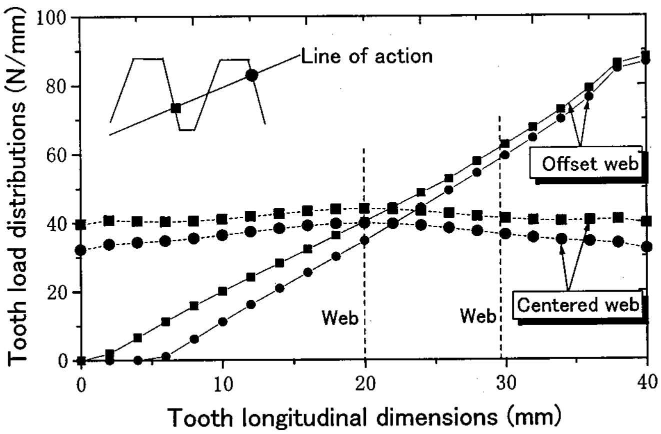


Fig. 7 Tooth load of two teeth engagement

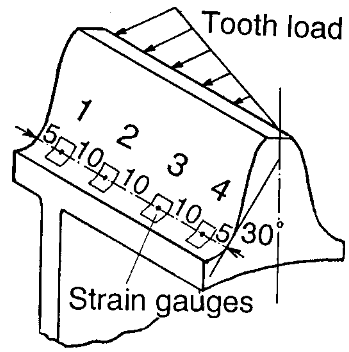


Fig. 8 Strain gauge positions in tooth root

direction of the line of action are calculated by FEM. This calculation is repeated for all the reference points from 1 to n and the deformation influence coefficient matrix $[a_{kj}]$ of gear① are constructed by these deformations. When to calculate the deformations, boundary nodes in the joint part of the web and hub as shown in Fig. 4(a) are fixed. The same calculations are done for gear② to get $[a_{k'j'}]$ [19].

4.5 Total Load P Along the Line of Action. For the gear transmission, external force **P** in Fig. 2 is equivalent to the total load **P** in the direction of the line of action. It can be calculated by Eq. (16).

$$\mathbf{P}=\mathbf{T}/r_g \tag{16}$$

Where, **T** is loading torque on one gear and $\mathbf{r}_g$ is the radius of the gear base circle. **T** is equal to 294 N.m in the paper.

4.6 Effect of Mesh Teeth Pairs. A gear usually has one tooth engagement position and two teeth engagement position when it rotates in one meshing period. This engagement position is given by a parameter in the program developed for the loaded tooth contact analysis. The program shall make a tooth contact model and divide the FEM meshes of the model automatically according to the parameter. Figure 5(a) is the simplified image of one tooth contact model only in the meshing teeth part and FEM mesh-dividing (Fig. 4 is the real model and mesh dividing). In Fig. 5(a), one reference face where there are a lot of meshes is made. Equations (11), (12) and (13) are built upon this face to calculate the load distributions (see Fig. 6).

If you want to calculate tooth load distributions shared by two pairs of teeth, the two teeth engagement position parameter is given to the program and the two teeth contact model and FEM mesh-dividing are made by the program automatically. Figure 5(b) is the imagine in this case. There are two reference faces in Fig. 5(b) and Eqs. (11), (12) and (13) are built up on the two faces at the same time, then load distributions on these two faces can be calculated at the same time (see Fig. 7). A program that can calculate 1～4 teeth engagements has been developed.

5 Calculation and Experiment Results

5.1 Tooth Load Distributions. 3D, tooth load distribution of one tooth engagement is shown in Fig. 6. It is found that contact width of the end close to the web is wider than the end far from the web. Tooth loads of two teeth engagement are summed to distribute on the geometrical contact line and shown in Fig. 7. It is the load distribution per unit length of the geometrical contact line. Tooth load distributions when the thin web is centered are also calculated to compare with the offset web structure in Fig. 7. It is found that web position has a great effect on tooth load distributions of the 3DTRG.

5.2 Tooth Strains and Contact Patterns. Strain gauges (gauge length = 0.2 mm, width = 0.9 mm) are pasted on the compressive side of the 3DTRG as shown in Fig. 8. Though strains on the tensile side of a tooth are more important than the compressive side for the tooth root strength design, they are easy to be affected by the friction on the meshing tooth surface. Also, gauges on the tensile side are easy to be touched by the tip of the mating gear. So, the compressive side of the tooth is used to paste the gauges. A power-circulating form gear test rig as shown in Fig. 9 is used for the test. A very low speed (1.65 rpm) is used in the test.

Tooth root strains at the position 1, 2, 3 and 4 as shown in Fig. 8 are measured and compared with calculated ones. Figure 10 is

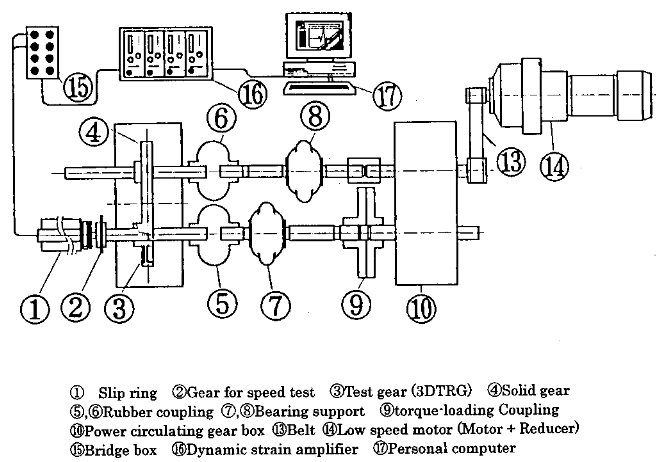


Fig. 9 Power-circulating test rig

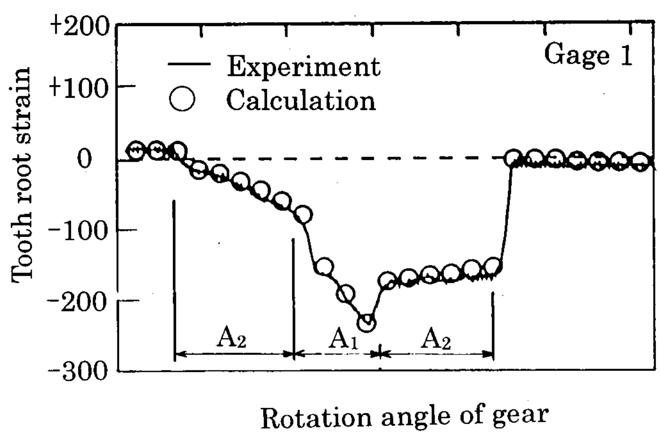
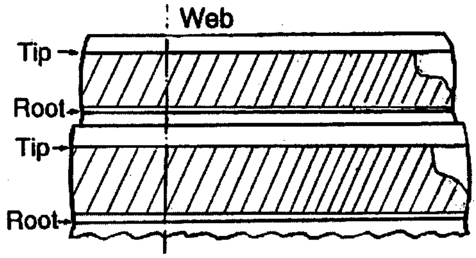
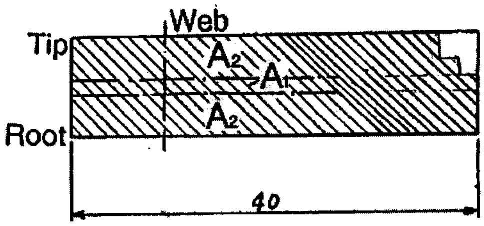


Fig. 10 Tooth root strains comparison



(a) Measured contact pattern



(b) Calculated contact pattern

Fig. 11 Tooth contact pattern comparison

the strain comparison at position 1. ○ is calculated strain and—is the measured one. A1 and A2 are the strain signals at one tooth engagement and two teeth engagement. From Fig. 10, it can be concluded that the calculated strain agrees with the measured one very well. This conclusion is also suitable for gauge position 2, 3, 4. Figure 11 is the comparison of tooth contact pattern. Thickness of the paint used to measure the tooth contact pattern is controlled not to be greater than 2 micron in the test.

6 Effects of Gear Models

6.1 Line-Contact and Face-Contact Model. The line-contact and the face-contact model are both used to calculate tooth loads for a pair of solid gears with the same gear parameters as shown in Table 2 and compared in Figure 12. It is found that the line-contact model has 5% difference with the face-contact model. This conclusion is also suitable for 3DTRG.

6.2 Partial Tooth FEM Model. Simplified image of a partial tooth deformation model (four-tooth model), usually used for the solid gears, is shown in Fig. 13. This four-tooth model is used to calculate the tooth loads under three kinds of boundary conditions as shown in Fig. 14. These loads are compared with the whole gear model in Fig. 14. From Fig. 14, if the rim, the web and the hub of the four-tooth model are fixed, the tooth load distribution is almost the same as the one with a pair of solid gears. If the rim of the four-tooth model is free, tooth load distribution is very close to the one with the whole gear model. This means that the rigidity of the thin rim has a substantial effect on the load distribution of the thin-rimmed gear. (Notes: Fig. 13 is only an imagine of the partial tooth deformation model, not the real mesh-dividing pattern. FEM meshes in Fig. 13 are very coarse in order to see clearly. The real FEM meshes are divided as fine as Fig. 4).

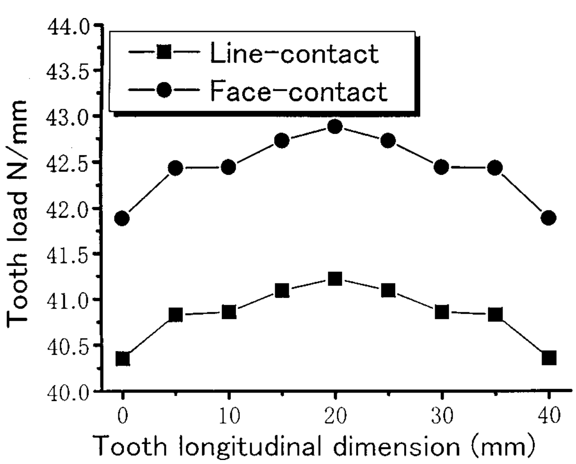


Fig. 12 Tooth load comparison

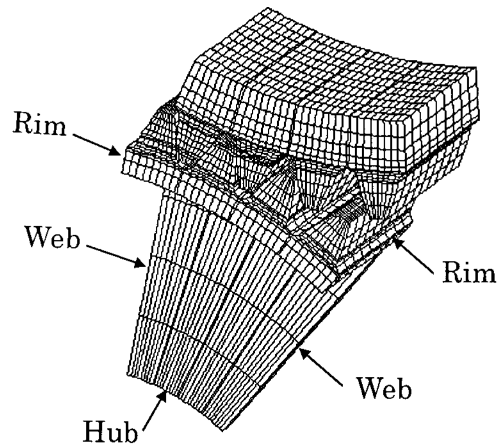


Fig. 13 Four-tooth FEM model

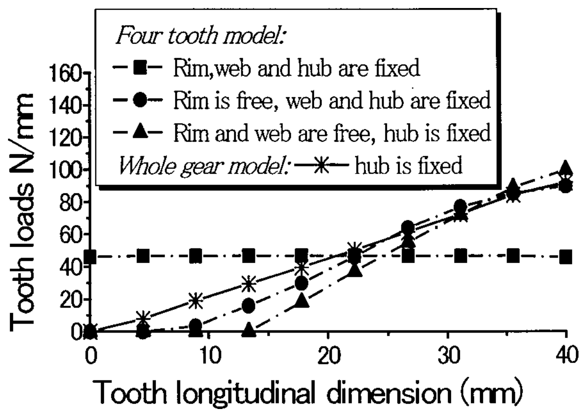


Fig. 14 Load distribution comparison

7 Conclusions

(1) The combination of the mathematical programming method with the three-dimensional, finite element method is successful in bending stress analysis for the thin-rimmed gears. A loaded tooth contact analysis program was developed to calculate all of the three-dimensional, thin-rimmed gear structures with all of the gear parameters, so it can be used as a tool for thin-rimmed gear strength design.

(2) A partial tooth model is not suitable for the thin-rimmed gears if the rim is fixed on the boundary.

(3) Line-contact model has about 5% difference from the face-contact model.

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